

Practical - 9

[page Replacement Policy]

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

INPUT :

```
GNU nano 8.7 optimal.c
#include <stdio.h>

int main() {
    int pages[50], frames[10];
    int n, f, i, j, k;
    int hit = 0, fault = 0;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &pages[i]);

    printf("Enter number of frames: ");
    scanf("%d", &f);

    for(i = 0; i < f; i++)
        frames[i] = -1;

    for(i = 0; i < n; i++) {
        int found = 0;

        for(j = 0; j < f; j++) {
            if(frames[j] == pages[i]) {
                hit++;
                found = 1;
                break;
            }
        }

        if(found == 0) {
            int pos = -1, farthest = i;

            for(j = 0; j < f; j++) {
                int k;
                for(k = i + 1; k < n; k++) {
                    if(frames[j] == pages[k])
                        break;
                }

                if(k == n) {
                    pos = j;
                    break;
                }

                if(k > farthest) {
                    farthest = k;
                    pos = j;
                }
            }

            if(pos == -1)
                pos = 0;

            frames[pos] = pages[i];
            fault++;
        }
    }

    printf("\nPage Hits = %d", hit);
    printf("\nPage Faults = %d", fault);
    printf("\nHit Percentage = %.2f%%", (float)hit/n*100);
    printf("\nFault Percentage = %.2f%%\n", (float)fault/n*100);

    return 0;
}
```

OUTPUT :

```
HP@Hiteshri MSYS ~  
$ nano optimal.c  
  
HP@Hiteshri MSYS ~  
$ gcc optimal.c -o optimal  
  
HP@Hiteshri MSYS ~  
$ ./optimal  
Enter number of pages: 12  
Enter page reference string:  
1 2 3 4 1 2 5 1 2 3 4 5  
Enter number of frames: 3  
  
Page Hits = 5  
Page Faults = 7  
Hit Percentage = 41.67%  
Fault Percentage = 58.33%
```